MATH 680–2026
Computation Intensive Statistics (4 credits)

Instructor: Archer Yi Yang
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Lecture: Tuesday, Thursday 11:35 am – 12:55 pm; Burnside 1205
Office Hour: Thursday 10:30 am – 11:30 am
Prerequisites: MATH 556, MATH 557 or permission of instructor

Course Website: the course website will be hosted on MyCourses

You will be able to access all online course material through this website. If you encounter any problems, please ask the course instructor.

References: optional, no required textbook:

1. The Elements of Statistical Learning (2nd Ed) by T. Hastie, R. Tibshirani and J. Friedman
2. Pattern Recognition and Machine Learning by Christopher Bishop
3. Deep Learning: Foundations and Concepts by Chris Bishop and Hugh Bishop

Links to other learning resources will be provided on MyCourses.

Software R (`http://www.r-project.org/`) and Python.

Audience The course will be aimed at first- and second-year students in the statistics/biostatistics master and Ph.D. program. Students from other departments or programs are welcome, space permitting; instructor permission required.

Grading

- **Class participation (10%):** There are participation mark taken at each lecture based on the questions and discussion.
- **Homework (30%):**
  - There will be five homeworks.
  - Each homework must be submitted through Mycourses. For theoretical questions, you can submit hand written answers. For computational questions, you must submit a typed written answers (e.g. LaTeX or Microsoft Word, here's a template).

  - Students can complete assignments **in groups,** each consists of **2 persons**. You need to **change your teammates** for each assignment and make sure that you only work with the same person once.

- **Team presentation (10%):** In teams of two, students will present a paper, explaining the statistical and machine learning methods involved and discussing its empirical findings.

- **Course project (40%** report + **10%** presentation**):**
  - There will be a course project with two milestones: a final report and a class conference.
  - Students may work individually or in teams of two. Each team will present its project at the class conference and submit a written technical report. The final report must clearly describe each student's individual contribution.
  - Each team must propose a project related to the course topics and **obtain approval from the instructor.**
  - The main report is limited to 10 pages and must use the template (`https://media.icml.cc/Conferences/ICML2026/Styles/icml2026.zip`); References and appendices do not count toward the 10-page limit and have no page limit. An additional project template is available

- **Exam:** none

Policy on collaboration: Collaboration on homework assignments with fellow students is encouraged. However, such collaboration should be clearly acknowledged, by listing the names of the students with whom you have had discussions concerning your solution.

Programming languages: No restriction.

Tentative Topics:

- Matrix decompositions – PCA, matrix completion, spectral clustering, Pagerank, word2vec
- Graphical representation learning, node2vec, graphical neural network etc
- Contrastive representation learning, SIMCLR, CLIP etc
- Causal representation learning
- Jackknife, Bootstrap
- MM algorithm, EM algorithm

- Variational inference, topic modelling
- Other topics, time permitting

Feedback: I welcome comments and suggestions (and complaints) and will solicit feedback via a survey partway through the class. Comments at any other time are welcome, and if you prefer anonymity, you can leave a note in my mailbox or under my door.

When to see me about an assignment: I'm here to help, including providing guidance on assignments. But before coming to see me about a difficulty, you should try something a few different ways and try to define/summarize what is going wrong or where you are getting stuck.

+ **In the event of extraordinary circumstances beyond the University's control, the content and/or evaluation scheme in this course is subject to change. If you need special assessment arrangements or accommodations, please contact the Office for Students with Disabilities at 514–398–6009.**

Assessments and Plagiarism: Any material (assignment solution scripts, R scripts etc.) that you hand in for assessment should be your work alone, unless the assignment concerned is a designated group project. This does not mean that you cannot consult or collaborate with others, merely that the material submitted for assessment must have been produced by you.

Shortlist of Resources: The Department of Mathematics and Statistics, as well as the University at large, have many resources to help you succeed in this course and throughout your degree. A more extensive list of resources can be found in the MyCourses Webpage for this Course (under "Content"), as well as the Department's EOSW webpage. A shortlist of the most commonly used resources is available here:

- The Wellness Hub is a centralized website for student physical and mental health resources.
- The Math Help Desk is staffed by knowledgeable math students who can help answer your questions related to your courses. They have tutors on M-F from noon-5 PM in Burnside Hall room 911.
- Statistics Online for Students is a resource which offers online help on statistics courses in the department. Click on the link above and you will be added to a Microsoft Teams group where the schedule will be announced.
- SUMS is the Society of Undergraduate Mathematics Students. Join their Facebook group to get on their listserv, connect with other students in the department, and participate in some of their activities/social events.

- GSAMS is the Graduate Student Association for Mathematics and Statistics (GSAMS). All graduate students are represented by GSAMS, and they hold regular events throughout the semester to promote graduate student community.
- Advising is an important resource to guide you throughout the duration of your degree, and advisors can help with answering questions related to your degree program. Check out the Department's Advising Website to find out how to get in touch with a Departmental Advisor, should you need to do so.
- The Office for Mediation and Reporting is a McGill centralized office used to file a formal report of discrimination, harassment, or sexual violence; learn about policies and processes; or be connected to additional supports.
- The Office for Sexual Violence, Response, Support and Education provides support for all members of the McGill community who have been impacted by sexual violence (whether it be sexual harassment or assault, gender-based or intimate partner violence, or cyberviolence) and works to foster a culture of consent on campus and beyond.
- The Office of the Dean of Students (Case Management) is a collaborative process between a student, a Case Manager, and other concerned parties with the intention of improving the student's academic and personal outcomes. Case managers are trained in confidentiality, disclosures of sexual violence, and mental health first aid.

MCGILL UNIVERSITY POLICY STATEMENTS

The following three statements are included in this course outline, in keeping with Senate resolutions:

1. *McGill University values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism and other academic offences under the* ***Code of Student Conduct and Disciplinary Procedures****. For more information, see*

   `www.mcgill.ca/students/srr/honest/`

   [*Approved by Senate on 29 January 2003*]

2. *In accord with McGill University's Charter of Students' Rights, students in this course have the right to submit in English or in French any written work that is to be graded.*

   [*Approved by Senate on 21 January 2009*]

3. *Instructors who may adopt the use of text-matching software to verify the originality of students' written course work must register for use of the software with Educational Technologies and must inform their students before the drop/add deadline, in writing, of the use of text-matching software in a course.*

   [*Approved by Senate on 1 December 2004*]

If you have a disability and need special arrangements, please contact the **Office for Students with Disabilities** at 514–398–6009.